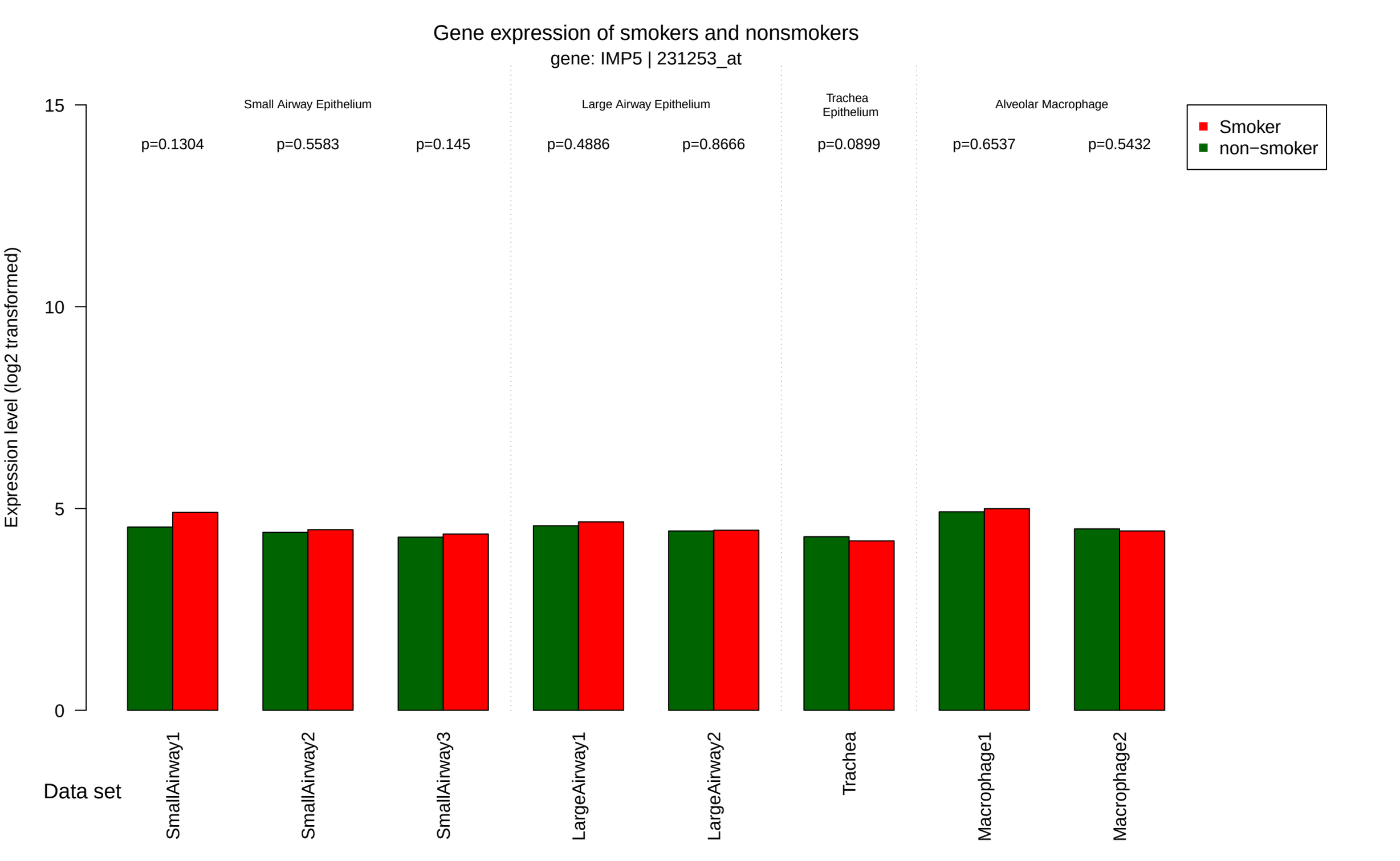
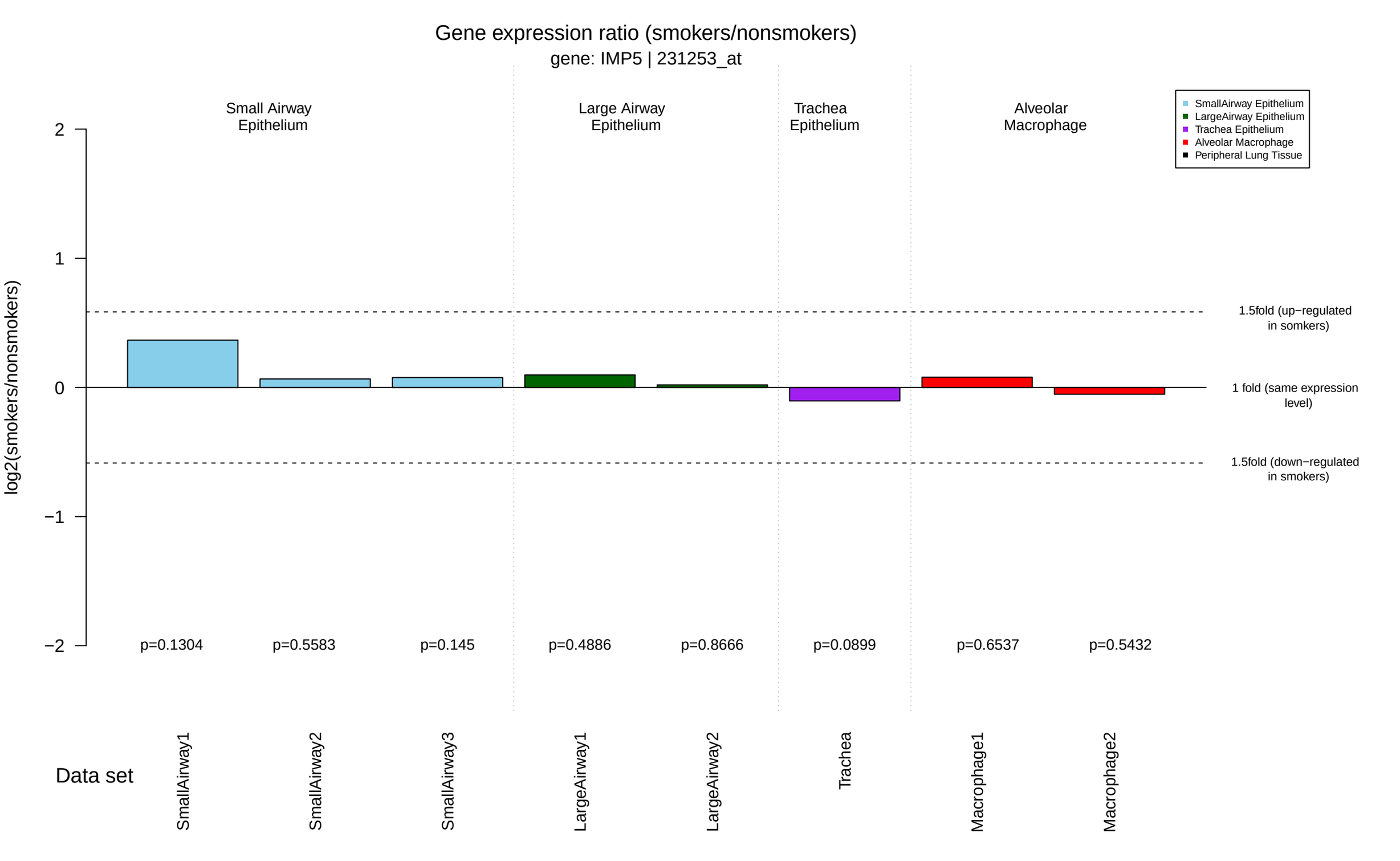
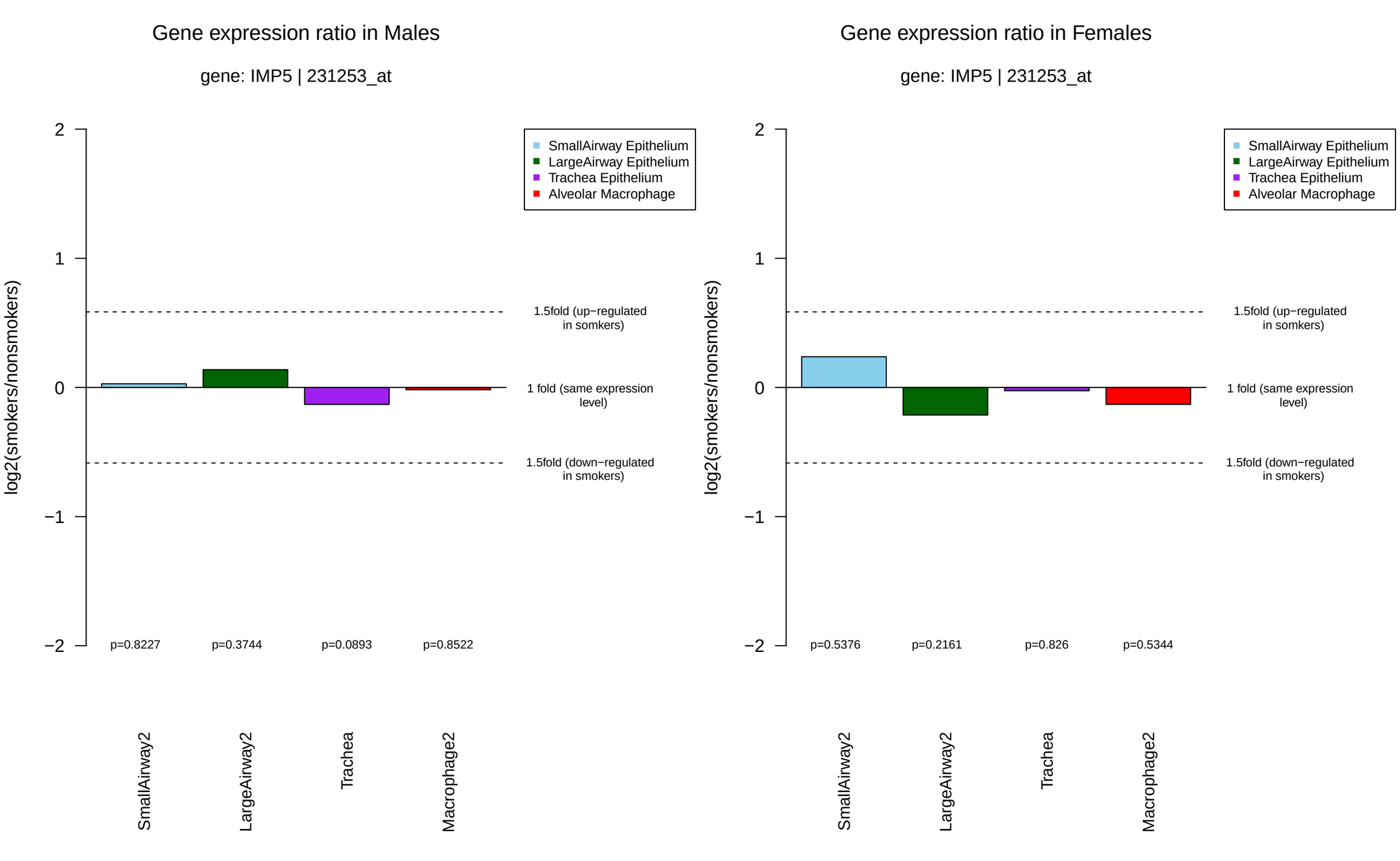
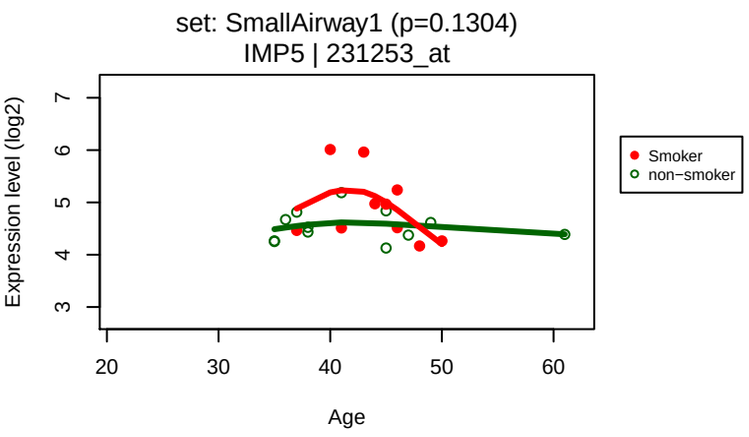
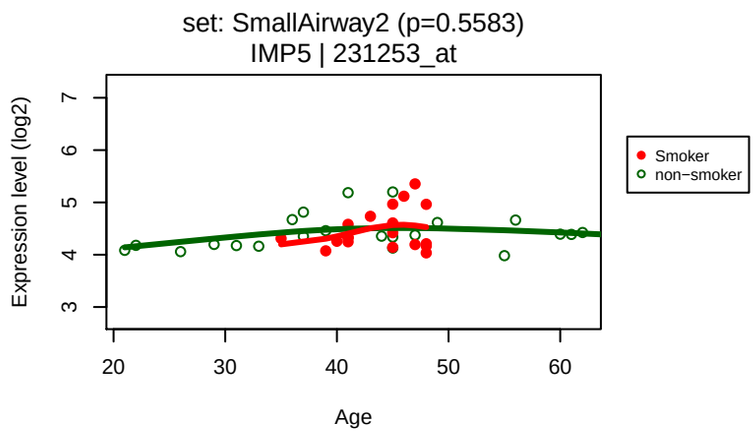
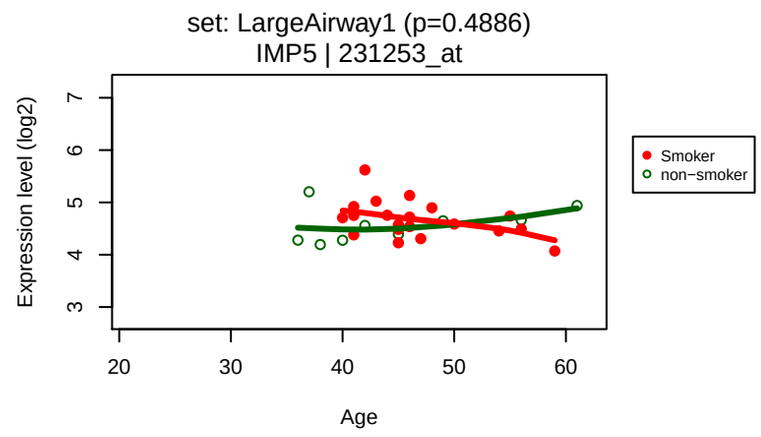

Gene expression of smokers and nonsmokers
gene: IMP5 | 231253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1304
p=0.5583
p=0.145
p=0.4886
p=0.8666
p=0.0899
p=0.6537
p=0.5432
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

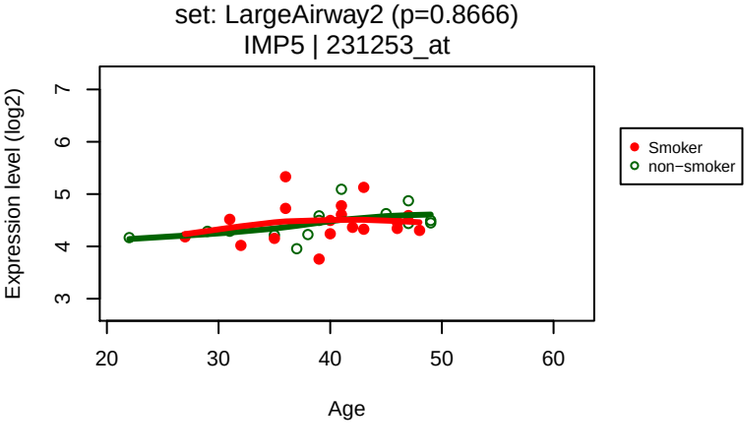

Gene expression ratio (smokers/nonsmokers)
gene: IMP5 | 231253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1304
p=0.5583
p=0.145
p=0.4886
p=0.8666
p=0.0899
p=0.6537
p=0.5432
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

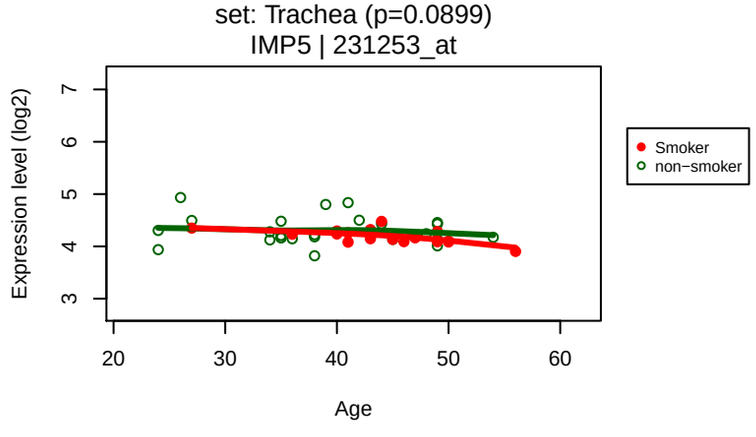

Gene expression ratio in Males
gene: IMP5 | 231253_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8227
p=0.3744
p=0.0893
p=0.8522
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMP5 | 231253_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5376
p=0.2161
p=0.826
p=0.5344
SmallAirway2
LargeAirway2
Trachea
Macrophage2

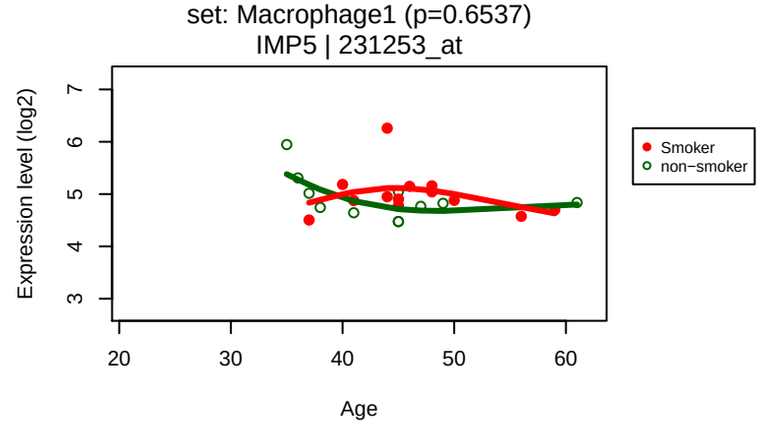

set: SmallAirway1 (p=0.1304)
IMP5 | 231253_at
Expression level (log2)
Age
Smoker
non−smoker

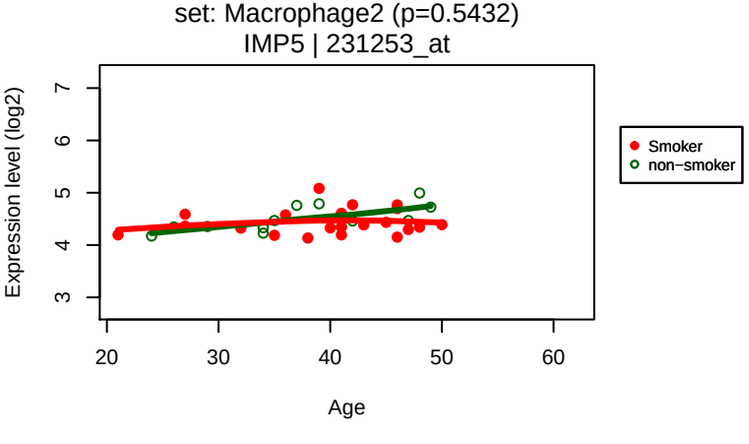

set: SmallAirway2 (p=0.5583)
IMP5 | 231253_at
Expression level (log2)
Age
Smoker
non−smoker

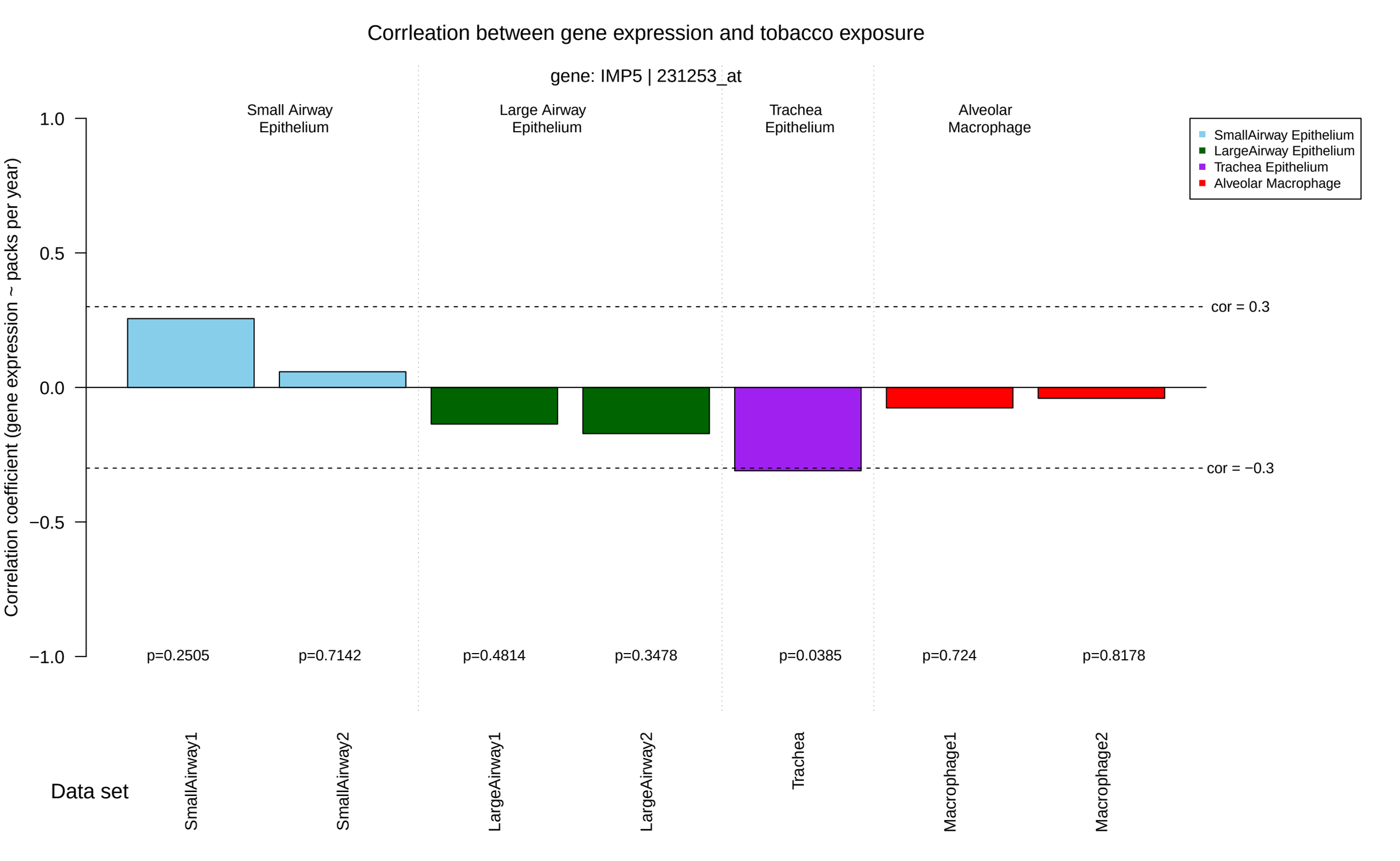

set: LargeAirway1 (p=0.4886)
IMP5 | 231253_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8666)
IMP5 | 231253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0899)
IMP5 | 231253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6537)
IMP5 | 231253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5432)
IMP5 | 231253_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMP5 | 231253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2505
p=0.7142
p=0.4814
p=0.3478
p=0.0385
p=0.724
p=0.8178
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2